

UrbanPulse Task A Report

Task A Report

Problem Statement

Provide the architecture view that explains how UrbanPulse handles multi-source telemetry and multi-layer analytics.

Design

Kafka is the shared ingestion plane; Flink handles real-time incident detection; Spark handles analytics and reporting; Parquet and Kafka sinks support storage and serving.

Implementation

The platform uses the architecture described in ``architecture/architecture.md`` and the supporting matrix in ``architecture/lambda_vs_kappa.md``.

Code Snippets

See ``create_topics.py``, ``flink/incident_detector.py``, and ``spark/ward_analytics.py``.

Screenshots

See ``screenshots/overview.svg``.

Results

The architecture report is now generated as both Markdown and PDF artifacts.

Challenges

Keeping the architecture file concrete while staying lightweight enough for local execution.

Future Work

Connect the topology to Grafana and a REST API in a production deployment.

Conclusion

Task A is now documented and packaged for submission.